Introduction

Electrical components are used in many important products that we interact with on a daily basis, such as consumer electronics like our phones, to industrial applications, and power generation. Accurate and fast identification of electrical components allows for better repairs, easier quality control, safer electronics disposal, and correct design and assembly of electrical parts. Using a trained machine learning model, we can reduce the amount of time needed to identify electrical components. We will be using a convolutional neural network because CNN can automatically extract meaningful features such as edges, shapes, textures, and contours through stacked convolutional layers.

There have been recent works regarding using AI to identify electrical components, including:

- Özkan & Dündar (2023): Used data augmentation and compared models such as Vision Transformer, MobileNetV2, EfficientNet, Swin Transformer, and DeiT on five component classes. They found DeiT to perform best, though the dataset was limited in component variety (UYSAL, F., 2025).
- IRJET (2024): Dataset contains 4,038 images of six component types and evaluates AlexNet, GoogleNet, ResNet50, and a custom CNN. They found ResNet50 to perform best. Results indicated that model performance could be improved through architectural tuning and hyperparameter optimization (Nath, J. et al., 2024).
- PMC (2022): Used a dataset containing seven component categories. Applied transfer learning with pretrained models (VGG16, ResNet50, InceptionV3), using resizing, filtering, and augmentation to enhance robustness. The paper highlights the need for greater dataset diversity and better real-time performance (Chand, P. et al., 2022).

Although existing research has achieved promising results, challenges remain in expanding component coverage. In this project, we aim to develop a more comprehensive model capable of accurately identifying a wider range of electrical components.

Dataset and Preprocessing

Dataset Overview

The dataset consists of 3 folders, train (1422 files), test (200 files) and valid (409 files). Each folder contains images and labels subfolders. Images are .npy files representing 640×640 RGB arrays of electrical components, while labels are .csv files corresponding to each image.

Data Processing

Each image was parsed to crop every object. Crops were converted to PIL images and saved in dataset_cropped. This new folder mirrors the original split with train, test, and valid subfolders.

Analysis of the cropped dataset showed 31 classes in train, 27 in test, and 26 in valid. While all valid and test classes appeared in train, several train classes were absent from both splits, which may reduce evaluation accuracy and limit the model's ability to generalise to unseen classes.

Data Imbalance Analysis

Further exploratory analysis of the cropped training set (4,896 samples) showed a 6.5× class imbalance, ranging from 51 samples (Class 15) to 333 (Class 30), with 8 classes in the low range (<91.5 samples) and 8 classes in the high range (>175.8 samples). Refer to Fig. 1.1.

To address significant class imbalance, we implemented class-weight balancing during training. This approach assigned higher weights to underrepresented classes (e.g., Class 15 with 51 samples received weights >2.0) and lower weights to overrepresented classes (e.g., Class 30 with 333 samples received weights <0.4), creating a strong inverse correlation between sample count and class weight. The rebalancing resulted in a more uniform distribution across all 32 classes, ensuring each class contributes more equally to the loss function (Refer to Fig. 1.2). During gradient updates, minority classes now exert proportionally larger influence, improving model fairness and generalisation by preventing bias toward dominant classes.

Data Augmentation Strategy

The training data was then augmented through random rotation, translation, horizontal and vertical flipping, as well as contrast adjustment. This expands training data, reduces overfitting, and improves robustness of the model to geometric and photometric variation of unseen data (Chauhan, Y., 2022).

Methodology

Transfer Learning Approach

We decided to adopt a transfer learning approach using pre-trained convolutional neural networks (CNNs), EfficientNetV2-S and ResNet50 for image classification.

Pre-trained models reduce data and computational requirements by leveraging learned features such as color

and shape, enabling faster and more efficient training on our custom dataset (Barman, R. et al., 2019).

In this project, we compare the performance of these two popular CNN architectures and select the better-performing model as our final classifier. To evaluate non-CNN performance, we also tested a Vision Transformer built from scratch and a pre-trained vit_b_16 from Hugging Face.

Model Architecture

ResNet50 is a deep CNN introduced in 2015 while EfficientNetV2-S is a CNN with Fused-MBConv blocks introduced in 2021. Also, both models are available in widely used deep learning libraries such as TensorFlow and Pytorch, making them practical choices for our implementation.

ResNet50: A key feature of ResNet50 is its use of residual connections, which are simple paths that allow information skip ahead instead of going through every layer in the model, making the learning more stable and efficient (Vina, A., 2025). We also chose to use ResNet50 as ResNet50 remains a standard baseline in image classification, offering interpretability and benchmark comparability. It is well-suited for datasets with clear, structured visual patterns (like electrical components). By using ResNet50, we can have a meaningful comparison against newer architectures like EfficientNetV2-S.

EfficientNetV2-S: On the other hand, with the Fused-MBConv blocks, it reduces the computational cost of EfficientNetV2-S compared to ResNet50 while maintaining similar representational power, making it more computationally efficient than ResNet50 (Emergent Mind, n.d.). Generally, the accuracy of EfficientNetV2-S is also higher than that of ResNet50 on ImageNet. We used EfficientNetV2-S as it balances depth, width, and image resolution, giving strong accuracy with fewer parameters. It includes advanced normalization and dropout schemes that naturally reduce overfitting. This is supported by our 3-fold CV showing smooth and stable loss decline across all folds with no divergence between train and validation curves.

Vision Transformer Experiments

To evaluate non-CNN performance, we also tested a Vision Transformer (ViT) built from scratch, guided by the official Keras documentation on image classification with ViTs (Khalid, S., 2021), as well as a pre-trained vit_b_16 from Hugging Face, implemented with reference to a GitHub repository demonstration on transfer learning with a ViT model pre-trained on ImageNet and applied to a custom dataset (Gutlapalli, N., 2023).

We decided to try using a ViT as they work by breaking images into patches, which could help the model capture small specific patterns present across multiple electronic components.

For the self-built ViT model, both training and validation accuracy remained below 0.4 percent and did not significantly increase or decrease throughout model training. We concluded that this was due to the relatively small size of our dataset, as ViTs are known to perform significantly worse on small datasets compared to large datasets.

We then attempted to apply transfer learning with the pre-trained ViT vit_b_16 from Hugging Face. This model had far better training accuracy but validation accuracy remained relatively low. The final model had a test loss of 4.4962 and a test accuracy of 0.2262.

We concluded that ViTs were not a good option for our dataset, both due the small size of our dataset and due to ViTs being more computationally intensive, especially for small datasets (Rustamy, F., 2023). We decided to focus on optimizing our CNN models as they are better and more efficient options for our dataset.

Training Strategy

We performed 3-fold stratified cross-validation (CV) on both EfficientNetV2-S and ResNet50. In each fold, two parts were used for training and one part for validation, with training conducted over 10 epochs per fold. We selected 10 epochs as it is a commonly used training duration that balances computational efficiency with sufficient model convergence for preliminary experimentation.

Class-weight balancing and data augmentation were applied consistently across all training runs. We used the Adam optimizer with categorical cross-entropy as the loss function and accuracy as the primary training metric.

EfficientNetV2-S and ResNet50 training and validation metrics across 10 epochs for 3 folds are shown in Table 1.

EfficientNetV2-S: Validation accuracy reaches above 80% by epoch 2 across all folds, with training and validation curves intersecting around epochs 3-7. After epoch 8, validation loss plateaus and slightly exceeds training loss, while train accuracy begins to diverge above validation accuracy, indicating potential overfitting. We conclude that training for approximately 8 epochs provides optimal balance between accuracy (~94%) and generalisation.

ResNet50: Similar convergence pattern to EfficientNetV2-S, with validation accuracy exceeding 80% by epoch 2-3. After epoch 6, validation loss plateaus, train accuracy is also higher than validation accuracy across all three folds, indicating possible overfitting. We conclude that training for approximately 6 epochs provides optimal balance between accuracy (~94%) and generalisation.

Results and Discussion

Model Evaluation Methodology

The performance of EfficientNetV2-S and ResNet50 was assessed using multiple metrics: filtered accuracy, weighted F1-score, and macro-averaged F1, precision, and recall.

Performance Comparison

In Fig. 2 and Table 2, while EfficientNetV2-S achieved higher filtered accuracy (92.95% vs. 91.15%) and weighted F1-score (0.924 vs. 0.909) than ResNet50, these metrics are biased towards majority classes. Given that all 32 electronic component classes hold equal practical importance, macro-averaged metrics provide a more appropriate evaluation framework.

ResNet50 demonstrated superior macro F1-score (0.763 vs. 0.746), macro precision (0.767 vs. 0.740), and macro recall (0.784 vs. 0.769), indicating more consistent performance across both majority and minority classes.

Model Selection Justification

ResNet50 was selected as our final model based on its balanced performance profile. The architecture's consistent performance across both majority and minority classes suggests better generalisation capabilities for component recognition tasks, where reliable identification of all component types is operationally critical.

Test Set Performance

The final ResNet50 model was evaluated on the held-out test dataset containing 669 images. Performance metrics of this final model are presented in Table 3.

The model achieved an overall accuracy of 90.98% and filtered accuracy of 91.74% on the test set, demonstrating strong generalisation to unseen data. These results are consistent with using validation dataset (89.50% and 91.15% respectively), indicating that the model does not overfit and maintains stable performance on new data.

For macro-averaged metrics, the test set results (macro F1: 0.735, macro precision: 0.730, macro recall: 0.758) are comparable to cross-validation performance (macro F1: 0.763, macro precision: 0.767, macro recall: 0.784). The slight decrease of approximately 2-4% is expected when evaluating on completely unseen data and remains within acceptable margins.

The model successfully made predictions for 659 out of 669 test samples, with only 10 missing predictions (1.5% of the test set). This indicates robust performance across the majority of component classes, though some underrepresented classes may require additional training data or targeted augmentation strategies.

Societal Considerations

A human who has experience working with electrical components may be able to identify the electrical components with higher accuracy, but they would have to take a significant amount of time to identify large numbers of electrical components and would also be a very tedious task. Our model is useful as it is able to identify components at a far faster rate than a human would be able to, making it very useful for large-scale classification of electrical components, such as for component verification in assembly of electronics.

Our project has potential applications in supporting quality control and maintenance workflows for electrical components. It could also aid engineers and manufacturers in supply chain management and component identification tasks.

References

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Appendix

Fig. 1.1. Training Set Class Distribution with Quartile Thresholds

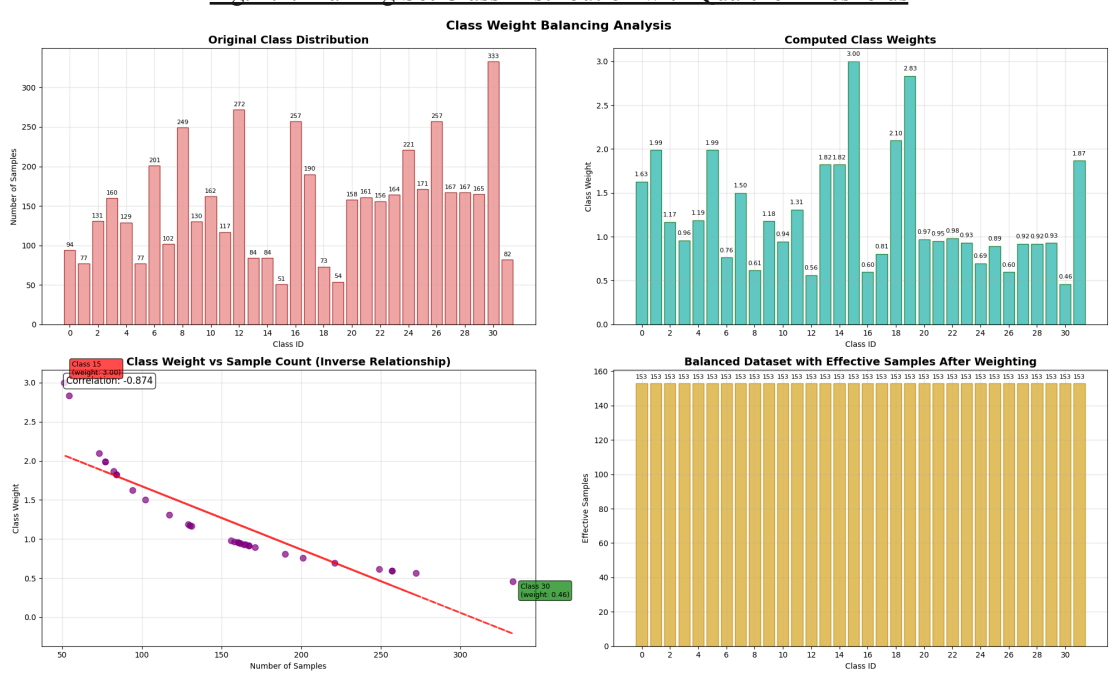


Fig. 1.2. Class Weight Balancing Analysis

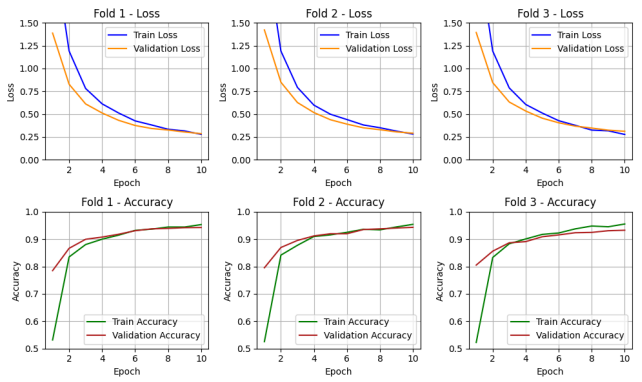
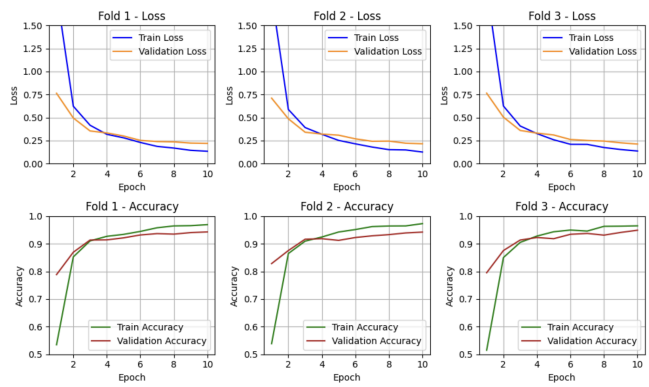
EfficientNetV2-S	ResNet50
	

Table 1. 3-Fold Cross Validation Comparison between EfficientNetV2-S and ResNet50

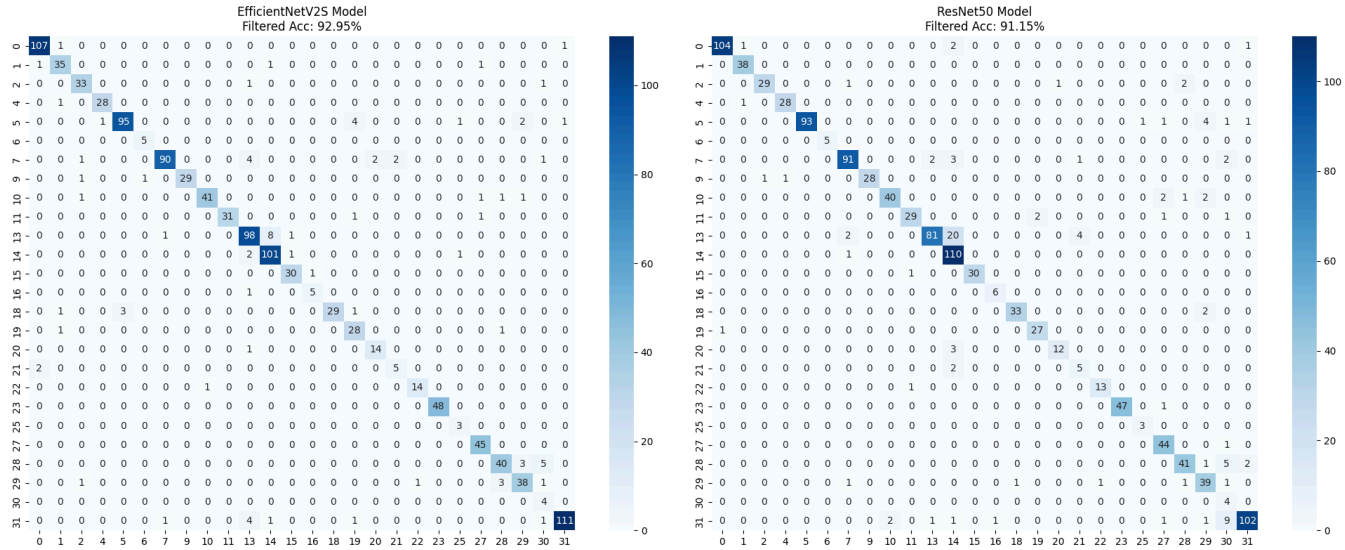


Fig. 2. Confusion-Matrix Heatmap Comparison (Rows: True Classes, Columns: Predicted Classes)

Metric	EfficientNetV2-S	ResNet50	Δ
Overall Accuracy	91.56%	89.50%	+2.07%
Filtered Accuracy	92.95%	91.15%	+1.79%
Macro F1-Score	0.746	0.763	-0.017
Weighted F1-Score	0.924	0.909	+0.016
Macro Precision	0.740	0.767	-0.027
Macro Recall	0.769	0.784	-0.016
Missing Predictions (samples)	18	22	-4

Table 2. Model Comparison using Performance Metrics

Metric	ResNet50
Overall Accuracy	90.98%
Filtered Accuracy	91.74%
Macro F1-Score	0.735
Weighted F1-Score	0.917
Macro Precision	0.730
Macro Recall	0.758
Missing Predictions (samples)	10

Table 3. Final Model Evaluation using Performance Metrics